

Propositional Logic Series

Episode 3: Worksheet Solutions

What Makes an Argument Valid?

Solutions

1. The counterexample criterion.

- (a) **Yes.** This is exactly a counterexample: every premise is true and the conclusion is false.
- (b) **No.** A counterexample must make all premises true together, not merely some of them.
- (c) **No.** A true conclusion does not supply the required true-premises/false-conclusion combination.
- (d) **No.** A false premise prevents the case from being a counterexample, even though the conclusion is false.

2. Reading possible cases.

Case C is a counterexample. It makes both premises true and the conclusion false. Cases B and D fail because one premise is false; Case A fails because the conclusion is true. Since Case C exists, the argument is **invalid**. One counterexample is enough.

3. A fixed mathematical argument.

If 18 is divisible by 6, then 18 is even.
18 is divisible by 6.
 $\therefore$ 18 is even.

- (a) **Valid.**
- (b) If both premises are true, the second premise makes the antecedent of the first premise true. The first premise then cannot be true while the conclusion is false. Therefore, no case makes both premises true and the conclusion false.
- (c) **Sound.** The argument is valid, and both premises are actually true: $18 = 6 \cdot 3$, and 18 is even.

4. A mathematical counterexample.

Invalid. The first premise is true because 5 is prime, so the disjunction has a true component. The second premise, that 5 is prime, is also true. The conclusion is false because 9 is not prime. Thus, the displayed facts themselves form a counterexample: all premises are true and the conclusion is false.

5. True statements are not enough.

- (a) **Yes.** When the card shows 12, it is divisible by 3, it is even, and it is greater than 10.
- (b) **No.** Actual truth in one case does not establish validity.
- (c) When the card shows 6, both premises remain true: 6 is divisible by 3 and even. The conclusion is false because 6 is not greater than 10. This case is a counterexample, so the argument is **invalid**.

6. Valid form with false content.

- (a) The first premise is false: 14 is a multiple of 7, but 14 is not odd. The second premise is true, and the conclusion is false.
- (b) **Valid.** The argument has the form $P \rightarrow Q, P, \therefore Q$. No case makes both premises true while making the conclusion false.
- (c) **Not sound.** Soundness requires validity and all actually true premises. The first premise is false.

7. Kinds of validity.

- (a) **Nomological.** The conclusion relies on the physical constraint that information cannot travel faster than light.
- (b) **Conceptual.** An ophthalmologist is a kind of doctor; the connection depends on the meanings of the concepts.
- (c) **Formal.** The argument has the pattern $P \vee Q, \neg Q, \therefore P$. Its validity does not depend on the particular occupations or person named.

8. Bivalence.

- (a) **Admissible.** Different propositions may receive different truth values.
- (b) **Not admissible.** Under bivalence, a proposition is not both true and false in the same case.
- (c) **Not admissible.** Under bivalence, each proposition under consideration is either true or false.
- (d) **Admissible.** Both propositions may be false.

9. Soundness.

- (a) **Sound.** Both defining conditions hold: validity and actually true premises.
- (b) **Not sound.** The true-premises condition fails.
- (c) **Not sound.** The validity condition fails. Actually true displayed statements do not repair invalidity.
- (d) **Not sound.** The validity condition fails. The false conclusion alone is not the reason for the classification.

10. Deductive does not mean valid.

If the device is fully charged, then the screen turns on.
 The screen turns on.
 $\therefore$ The device is fully charged.

- (a) **No.** Deductive describes the necessity claim being made; validity asks whether that claim succeeds.
- (b) Suppose the device is not fully charged, but an external power source turns the screen on. The conditional premise is true because its antecedent is false. The second premise is true because the screen is on. The conclusion is false because the device is not fully charged. This is a counterexample, so the argument is **invalid**.
- (c) The method is:
 1. identify all premises and the conclusion;
 2. try to make the conclusion false;
 3. keep all premises true at the same time;
 4. if such a case exists, stop and classify the argument as invalid.

Sources and Note

The conceptual background for these solutions is informed by P. D. Magnus, Tim Button, Robert Trueman, and Richard Zach, *forall x: Calgary: An Introduction to Formal Logic*, with contributions by J. Robert Loftis and Aaron Thomas-Bolduc, Chapter 2.

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